What a Balanced Affinity Sees: Sinkhorn Normalisation as a Helmholtz Projection of Directed Costs

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Abstract

Many datasets come with a direction at each point: RNA velocity, a gradient field, the velocity of a generative model. A natural way to let a neighbour embedding use it is to replace the symmetric ground cost by a direction-dependent one, such as a Randers metric, and keep the usual entropic affinity. We ask which part of the direction survives the normalisation of that affinity. For finite samples, we show that a Sinkhorn-balanced affinity is determined by the cost modulo separable functions. Its asymmetry therefore carries exactly the cyclic, non-gradient part of the cost's asymmetry, and a gradient drift, including a uniform one, is pure gauge. In the continuum limit, the Markov chain of a balanced affinity transports mass along a field that is divergence-free with respect to the sampling density. For the Laplace–Randers kernel this field has a closed form: at small drift it is the density-weighted Helmholtz projection of the drift onto its solenoidal part, whereas row-stochastic and diffusion-map normalisations transport along the whole drift. Two ingredients of practical affinities break the gauge. Per-row bandwidth calibration leaks gradient drifts at first order, and strongly; a squared cost leaks them at third order in the limit, though at first order with a small constant at the sample sizes we can simulate. Numerical experiments confirm the other predictions quantitatively, and Hodge decompositions of real RNA-velocity fields show when balancing helps and when it discards the signal.

1. Introduction

Neighbour embeddings such as SNE, t-SNE and UMAP (Hinton & Roweis, 2002; van der Maaten & Hinton, 2008; McInnes et al., 2018) build an affinity matrix from a symmetric ground cost and fit a low-dimensional layout whose own affinities match it. Doubly stochastic affinities, computed by Sinkhorn iterations (Sinkhorn & Knopp, 1967; Cuturi, 2013), make every point send and receive the same mass. SNEkhorn (Van Assel et al., 2023) showed that such marginals, in a symmetric entropic affinity, fix the hub and density problems of row-normalised affinities.

Many datasets, however, come with a *direction* at each point: RNA velocity in single-cell transcriptomics (La Manno et al., 2018; Bergen et al., 2020), the gradient of a loss with respect to activations, the velocity field of a flow-matching model, the displacement of an optimal-transport plan. A symmetric cost cannot express "cheaper to move with the flow than against it". The natural fix is a Finsler cost, and the simplest Finsler cost is the Randers metric (Randers, 1941; Bao et al., 2000):

$$F_x(v) = \|v\| + \omega(x)^\top v, \qquad \|\omega\| < 1. \tag{1}$$

Plugging such a cost into an entropic affinity is straightforward. It is much less clear what the resulting affinity actually encodes. The normalisation step rescales rows and columns, and rescaling can absorb part of the asymmetry. Indeed, in our experiments a Sinkhorn-balanced Randers affinity was nearly blind to some drifts and sensitive to others of the same magnitude.

This paper characterises that behaviour exactly. Our answer is a Helmholtz decomposition: *a balanced affinity sees the circulation of a drift and not its gradient part.*

Contributions.

- **Finite samples (Section 3).** A balanced coupling identifies its cost exactly modulo separable functions $f_i + g_j$ (Theorem 3.1). Consequently, the cyclic part of $\log(P_{ij}/P_{ji})$ equals $-2/\varepsilon$ times the cyclic part of the cost's antisymmetric part, for any diagonal-scaling normalisation. What differs between normalisations is only how gradient drifts reach the marginals, and balancing removes them. A uniform drift is invisible. Independently of the cost, the net mass flow of a balanced affinity is a pure circulation: it makes no net progress along any potential (Proposition 3.4).

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- **Continuum limit (Section 4).** Under a standard small-noise ansatz for the Sinkhorn potentials, the transport field of a balanced affinity is divergence-free with respect to the sampling density p and solves a strictly convex variational problem (Theorem 4.2). For the Laplace–Randers kernel it has a closed form; at small drift it is $-(d+1)\,\Pi_p\,\omega$, the p-weighted Leray projection. Row-stochastic and diffusion-map normalisations instead transport along the centroid field of the Randers unit ball (Debaussart-Joniec et al., 2026), which contains the whole drift.
- **Leakage (Section 5).** Practical affinities use a squared cost and per-row bandwidths. Per-row bandwidths leak gradient drifts at first order, through the part of $\nabla\phi/\sigma$ that is not a gradient. In the limit, the squared cost leaks them only at third order in their magnitude.
- **Evidence (Sections 6 and 7).** Each prediction is checked numerically, and Hodge decompositions of real velocity fields explain when a balanced affinity helps (cell-cycle circulation) and when it discards most of the signal (developmental trajectories).

2. Setting

Affinities are diagonal scalings of a kernel. Given points $x_1, \dots, x_n$ and a cost matrix $C \in \mathbb{R}^{n\times n}$, not necessarily symmetric, write $K = \exp(-C/\varepsilon)$ entrywise. The three normalisations we consider all rescale the rows and the columns of K,

$$\log P_{ij} = -C_{ij}/\varepsilon + \log\alpha_i + \log\beta_j, \tag{2}$$

that is $P = \operatorname{diag}(\alpha) K \operatorname{diag}(\beta)$, and differ only in how $\alpha, \beta > 0$ are chosen.

- *Row-stochastic* (RS), as in SNE's conditional affinities: $\beta = \mathbf{1}$ and $\alpha_i = 1/\big(n\,(K\mathbf{1})_i\big)$. Each row is a softmax of $-C_{i\cdot}/\varepsilon$ and $P\mathbf{1} = \mathbf{1}/n$.
- *Diffusion map* with exponent $\theta \in [0,1]$ (Coifman & Lafon, 2006): with degrees $d = K\mathbf{1}$, first $K \leftarrow \operatorname{diag}(d)^{-\theta} K \operatorname{diag}(d)^{-\theta}$, then row normalisation. The row factor $d_i^{-\theta}$ cancels, leaving $\beta_j = d_j^{-\theta}$ and $\alpha_i = 1/\big(n\,(K\beta)_i\big)$: a column factor fixed by the degrees, not by a prescribed marginal.
- *Balanced* (Sk): $\alpha = u$ and $\beta = v$ are the positive vectors, unique up to $(u,v) \mapsto (cu, v/c)$, with $u \odot Kv = a$ and $v \odot K^\top u = b$ for prescribed marginals a, b of equal mass, found by Sinkhorn iterations, here with $a = b = \mathbf{1}/n$.

SNE and t-SNE also calibrate a per-row bandwidth ε_i to a target perplexity (Vladymyrov & Carreira-Perpiñán, 2013). That divides row i of C by ε_i, which is not of the form (2); we treat it separately in Section 5. SNEkhorn (Van Assel et al., 2023) is *not* Sk. It minimises $\langle P, C\rangle$ under unit row sums, a per-row entropy floor and the constraint $P = P^\top$, with solution $P_{ij} = \exp\big((\mu_i + \mu_j - 2C_{ij})/(\gamma_i + \gamma_j)\big)$. For symmetric P, $\langle P, C\rangle = \langle P, \frac{1}{2}(C + C^\top)\rangle$, so SNEkhorn sees only the symmetric part of a cost and discards every drift, cyclic or not. Sk drops the symmetry constraint and keeps two free scalings $u \neq v$. For a symmetric C and equal γ_i the two coincide.

Randers costs. A Randers norm (Randers, 1941; Bao et al., 2000) adds a linear drift term to the Euclidean norm, $F_x(v) = \|v\| + \omega(x)^\top v$. It is positive and convex in v if and only if $\|\omega(x)\| < 1$, and it is asymmetric: moving along ω costs $1 + \|\omega\|$ per unit length, moving against it $1 - \|\omega\|$. A correctly fitted ω thus points *away* from the direction the affinity favours. The Randers distance $d_F(x \to y)$ is the infimum of $\int_0^1 F_{\gamma(t)}(\dot\gamma)\,dt$ over paths from x to y. It has no closed form when ω varies, for three reasons (Appendix A). In low dimension, $d_F(x_0 \to \cdot)$ solves an eikonal equation, which fast-marching schemes for asymmetric metrics handle on a grid (Sethian & Vladimirsky, 2003; Mirebeau, 2014), one source at a time. In the 10 to 50 dimensions of single-cell data a grid is out of reach, and graph shortest paths (Tenenbaum et al., 2000) need exponentially many points to converge. Above all, ω is measured only at the samples, so the infimum over paths through empty regions depends on an extension of ω the data do not support. We therefore use costs built from the endpoint values $\omega_i = \omega(x_i)$ only, with $\Delta_{ij} = x_j - x_i$:

- *frozen,* $\hat d(x_i \to x_j) = \|\Delta_{ij}\| + \omega_i^\top \Delta_{ij}$;
- *averaged,* $\bar d(x_i \to x_j) = \|\Delta_{ij}\| + \frac{1}{2}(\omega_i + \omega_j)^\top \Delta_{ij}$.

Both approximate the cost of the straight segment by a quadrature rule, left-endpoint and trapezoid, with errors $O(\|\nabla\omega\| r^2)$ and $O(\|\nabla^2\omega\| r^3)$ at $r = \|\Delta_{ij}\|$. Bending the segment gains only $O(\|\nabla\omega - \nabla\omega^\top\| r^3)$ (Appendix A). A kernel at bandwidth ε only weighs pairs with $r = O(\varepsilon)$, so these errors are small where the affinity looks. The two closed forms have the *same antisymmetric part* $\frac{1}{2}(\omega_i + \omega_j)^\top \Delta_{ij}$. They differ only in their symmetric part, by $\frac{1}{2}(\omega_i - \omega_j)^\top \Delta_{ij}$.

One case is exact. For a *gradient drift* $\omega = \nabla\phi$, the drift term integrates to $\phi(y) - \phi(x)$ along any path, so straight lines are geodesics and $d_F(x \to y) = \|y - x\| + \phi(y) - \phi(x)$. No convexity of ϕ is needed. Admissibility only asks $\|\nabla\phi\| < 1$, which makes ϕ 1-Lipschitz and d_F positive off the diagonal. Convexity would only order the costs: it puts the frozen cost below the exact one, since $\nabla\phi(x_i)^\top \Delta_{ij} \le \phi_j - \phi_i$. The averaged form's drift term is then a trapezoid rule for $\phi_j - \phi_i$, with error $O(\|\nabla^3\phi\|\, r^3)$.

The gauge: separable matrices. By (2), every normalisation adds to $-C/\varepsilon$ a matrix of the form $f_i + g_j$. Let

$$\mathcal{S} = \{(f_i + g_j)_{ij} : f, g \in \mathbb{R}^n\}.$$

If two costs differ by an element of $\mathcal{S}$, their kernels differ by diagonal factors on the left and right. A normalisation that chooses both scalings itself, such as Sk, absorbs those factors and returns the same P. So $\mathcal{S}$ is the set of cost changes that balancing cannot see, a gauge group in the physicist's sense. Theorem 3.1 shows that it is the *only* such set. To read it in terms of drifts, we need to know which asymmetric costs lie in $\mathcal{S}$.

Lemma 2.1 (Structure of $\mathcal{S}$)**.** *Let $\mathcal{A}$ be the antisymmetric $n \times n$ matrices and $\mathcal{G} = \{(\psi_j - \psi_i)_{ij} : \psi \in \mathbb{R}^n\} \subset \mathcal{A}$. Then*

$$\mathcal{S} = \{(s_i + s_j)_{ij} : s \in \mathbb{R}^n\} \oplus \mathcal{G}, \qquad \mathcal{S} \cap \mathcal{A} = \mathcal{G}. \quad (3)$$

with $\oplus$ a direct sum of subspaces.

Proof. Write $f_i + g_j = (s_i + s_j) + (\psi_j - \psi_i)$ with $s = \frac{1}{2}(f+g)$ and $\psi = \frac{1}{2}(g-f)$. The first term is symmetric and the second antisymmetric, so the sum is direct. If $f_i + g_j$ is antisymmetric, its diagonal gives $g = -f$, hence $f_i + g_j = f_i - f_j \in \mathcal{G}$. □

The symmetric half, $s_i + s_j$, is a per-point offset, such as a cost of leaving and entering each point; it carries no direction. The antisymmetric half is exactly the gradient flows. So among asymmetric costs, those balancing cannot see are precisely the differences of a potential, $\psi_j - \psi_i$, and nothing else. A uniform drift $\omega \equiv c$ is one: $c^\top \Delta_{ij} = \psi_j - \psi_i$ with $\psi = c^\top x$.

Hodge decomposition: what balancing keeps. Write $C = S + A$ with S symmetric and $A_{ij} = \frac{1}{2}(C_{ij} - C_{ji})$, the extra cost of going $i \to j$ rather than $j \to i$, an edge flow on the complete graph (Jiang et al., 2011; Lim, 2020). By Lemma 2.1 and Theorem 3.1, balancing cannot distinguish A from any $A + G$, $G \in \mathcal{G}$: it sees only the *class* $A + \mathcal{G}$. We need a canonical split of A into the part that is lost and the part that is kept. Given symmetric positive edge weights W, the weighted cyclic part of $B \in \mathcal{A}$ is

$$\mathrm{cyc}_W(B) = B - \arg\min_{G \in \mathcal{G}} \textstyle\sum_{ij} W_{ij}(B_{ij} - G_{ij})^2. \quad (4)$$

Adding $G \in \mathcal{G}$ to B shifts the minimiser by G, so $\mathrm{cyc}_W(B+G) = \mathrm{cyc}_W(B)$. Conversely, equal cyclic parts force $B - B'$ to be a difference of minimisers, a gradient. Hence $\mathrm{cyc}_W(B) = \mathrm{cyc}_W(B') \iff B - B' \in \mathcal{G}$: the cyclic part is a complete invariant of what balancing sees, for every W, and W only chooses the representative. The normal equations of (4) give $\sum_j W_{ij} \mathrm{cyc}_W(B)_{ij} = 0$ for every i: $\mathrm{cyc}_W(B)$ is the unique element of $B + \mathcal{G}$ with no net W-flow out of any point.

- A *gradient* flow $\psi_j - \psi_i$ is an asymmetry fully explained by a scalar ranking, such as a pseudotime. Moving up costs more because the target is higher. Its sum around every closed loop telescopes to zero.

- The *cyclic* part is what no ranking explains. Loop sums $B_{i_1 i_2} + B_{i_2 i_3} + \cdots + B_{i_m i_1}$ are unchanged by adding a gradient, so they are the observables of the class, and $\mathrm{cyc}_W(B)$ is the divergence-free flow that carries them. For a Randers drift term, the loop sum around a small triangle is $\oint \omega \cdot dx$, about the curl of ω times the area by Stokes' theorem. The cyclic part is thus a discrete curl: rotation around the cell cycle, or around any loop of states.

This is the discrete counterpart of the Helmholtz decomposition $\omega = \nabla\phi + \omega_{\mathrm{sol}}$; in the continuum limit W becomes the kernel weight times the density, and cyc_W becomes a density-weighted Leray projection (Section 4). On the k-nearest-neighbour graphs used on data, the cyclic part splits further into a local curl around triangles and a harmonic flow around holes of the graph (Jiang et al., 2011); the cell cycle is such a hole.

3. Finite samples: balanced affinities are gauge invariant

Theorem 3.1 (Identifiability modulo separable costs)**.** *Fix positive marginals a, b with equal mass and $\varepsilon > 0$. For a cost C, let $\mathrm{Sk}(C)$ denote the unique coupling of the form $\mathrm{diag}(u)e^{-C/\varepsilon}\,\mathrm{diag}(v)$ with marginals a, b. Then*

$$\mathrm{Sk}(C) = \mathrm{Sk}(C') \iff C - C' \in \mathcal{S}.$$

In particular, writing $C = S + A$ with S symmetric and A antisymmetric, $\mathrm{Sk}(S + A) = \mathrm{Sk}(S + A')$ if and only if $A - A' \in \mathcal{G}$.

Proof. The kernel is entrywise positive, so the scaling exists and the coupling is unique (Sinkhorn & Knopp, 1967). If $C'_{ij} = C_{ij} + f_i + g_j$, then $e^{-C'/\varepsilon} = \mathrm{diag}(e^{-f/\varepsilon})e^{-C/\varepsilon}\,\mathrm{diag}(e^{-g/\varepsilon})$. The diagonal factors are absorbed into u, v, so uniqueness gives the same coupling. Conversely, if both couplings equal P, then $\log P_{ij} = \log u_i + \log v_j - C_{ij}/\varepsilon = \log u'_i + \log v'_j - C'_{ij}/\varepsilon$. Hence $C - C' \in \mathcal{S}$. The second statement follows from Lemma 2.1. □

The second statement says that, among all asymmetric information, a balanced affinity retains exactly the cost's antisymmetric part modulo gradients. The asymmetry of the coupling itself makes this explicit.

An optimal transport reading. $\mathrm{Sk}(C)$ is the entropic optimal transport plan between a and b for the ground cost C (Cuturi, 2013; Peyré & Cuturi, 2019); with $a = b$ it is an entropic self-transport. The gauge $\mathcal{S}$ is then a familiar freedom of OT. On couplings $P \in \Pi(a, b)$, $\sum_{ij} P_{ij}(C_{ij}+f_i+g_j) = \langle P, C\rangle + \langle a, f\rangle + \langle b, g\rangle$, so a separable change shifts every transport cost by the same constant and leaves the optimal plans unchanged, with or without entropy (Villani, 2009; Santambrogio, 2015). Combined with the classical fact that adding a closed one-form to a Randers metric does not change its geodesics (Bao et al., 2000), this gives the continuous statement. For an exact drift $\omega = \nabla\phi$ the Randers cost is $\|y-x\|+\phi(y)-\phi(x)$, so unregularised OT under this cost between any μ and ν has the optimal plans of the Euclidean W_1 problem, and its value is $W_1(\mu, \nu) + \int \phi \, d(\nu - \mu)$. For self-transport the value is unchanged too. We have not found this stated, but it follows directly from the two facts above. Three points are specific to our setting.

- *The converse needs entropy.* Optimality of an unregularised plan is characterised by c-cyclical monotonicity, which involves the cost only through alternating cycle sums, and many costs share the same optimal plan. An entropic plan determines $\log P$, hence C up to $\mathcal{S}$ (Theorem 3.1). This is what makes the cyclic part $\mathrm{cyc}_W(A)$ the exact content of a balanced affinity.
- *The invariance is tied to the linear cost.* The squared Randers cost $(\|\Delta_{ij}\| + \phi_j - \phi_i)^2$ contains the cross term $2\|\Delta_{ij}\|(\phi_j - \phi_i)$, which is not separable. This is the source of the squared cost's leak in Section 5.
- *The normalisations are OT problems with different marginal constraints.* RS is entropic OT with only the row marginal imposed, a semi-relaxed problem. Leaving the column marginal free is what lets the gradient part through (see below). Unbalanced OT, with KL-penalised marginals (Chizat et al., 2018), would sit between RS and Sk and should see the gradient part with a strength set by the penalty. We leave this to future work.

Corollary 3.2 (Cyclic content of any diagonal normalisation)**.** *Let $P = \mathrm{diag}(\alpha)e^{-C/\varepsilon}\,\mathrm{diag}(\beta)$ with $\alpha, \beta > 0$. This covers row-stochastic, diffusion-map and balanced affinities. Then*

$$\log \frac{P_{ij}}{P_{ji}} = -\frac{2}{\varepsilon}A_{ij} + (\chi_j - \chi_i), \qquad \chi = \log\beta - \log\alpha,$$

and therefore $\mathrm{cyc}_W\big(\log P - \log P^\top\big) = -\frac{2}{\varepsilon}\,\mathrm{cyc}_W(A)$ for every weight matrix W.

The cyclic part of the asymmetry is thus fixed by the cost alone, and is the same for every normalisation. The normalisations differ in the gradient part χ, which is set by the marginals, and in what the marginals themselves are.

Corollary 3.3 (Gradient drifts are pure gauge)**.** *Let $\omega = \nabla\phi$ with $\|\nabla\phi\| < 1$. Take the exact Randers cost $C_{ij} = \|x_j - x_i\| + \phi(x_j) - \phi(x_i)$ and a single global ε. Then $\mathrm{Sk}(C) = \mathrm{Sk}(D)$, where $D_{ij} = \|x_j - x_i\|$, for every choice of marginals.*

- *A* uniform *drift $\omega \equiv c$ is the gradient of $c^\top x$, so a balanced affinity cannot detect it at all.*
- *Under the averaged closed form, the cyclic part of the drift term is $O(\|\nabla^3\phi\|\, r^3)$ on pairs at distance r. It vanishes for quadratic ϕ.*

Theorem 3.1 concerns log-ratios. A second, even simpler fact concerns the mass that actually moves, and it holds for *every* cost.

Proposition 3.4 (Balanced net flow is a circulation)**.** *Let P have equal marginals, $P\mathbf{1} = P^\top\mathbf{1} = a$. Then the net flow $N = P - P^\top$ is divergence-free at every point, $N\mathbf{1} = 0$. Equivalently, $N \perp \mathcal{G}$, and for every $f \in \mathbb{R}^n$*

$$\textstyle\sum_{ij} P_{ij}\,(f_j - f_i) = 0.$$

The same holds for every power of the chain $\mathrm{diag}(a)^{-1}P$.

Proof. $\sum_{ij} P_{ij}(f_j - f_i) = \sum_j a_j f_j - \sum_i a_i f_i = 0$. Taking $f = \mathbf{1}_k$ gives $(N\mathbf{1})_k = 0$. For gradient flows, $\langle N, (\psi_j - \psi_i)\rangle = \sum_{ij} N_{ij}(\psi_j - \psi_i) = -2\sum_i \psi_i (N\mathbf{1})_i = 0$. The chain $\mathrm{diag}(a)^{-1}P$ has stationary law a, and so do its powers. □

So a balanced affinity makes *no net progress along any potential*, for instance along a pseudotime, in any number of steps. It can only circulate. Progress around a cycle is still possible, because a phase increment is not the difference of a single-valued function. On a trajectory with no return path, any local forward motion must be compensated elsewhere, by mass sent backwards. In this language, Corollary 3.3 says that for a gradient drift the circulation is exactly that of the symmetric affinity, and Section 5 says how much spurious circulation practical affinities add.

For row-stochastic affinities the same gradient drift is *not* invisible. With $C = D + G$ and $G_{ij} = \phi_j - \phi_i$,

$$\mathrm{RS}(C)_{ij} \propto e^{-D_{ij}/\varepsilon}\, e^{-\phi_j/\varepsilon}.$$

This is the symmetric affinity with targets reweighted towards low ϕ, which is the direction of $-\omega$. The log-asymmetry is still a pure gradient (Corollary 3.2), so the information sits entirely in the column marginal, i.e. in where mass is sent. Balancing prescribes that marginal and so removes the drift. Section 4 turns this into a statement about transport.

Remark 3.5 (Two readings of a drift). Write a drift as a gradient part plus a circulating part. The gradient part says *where mass should go*; a normalisation that leaves column sums free records it as a marginal tilt. The circulating part says *how mass moves at fixed marginals*, and every diagonal normalisation records it identically. This is the dual of the snapshot non-identifiability of Weinreb et al. (2018): population snapshots constrain the potential part of a dynamics and not its rotation, whereas a balanced velocity affinity keeps the rotation and discards the potential.

4. Continuum limit: the balanced transport field

Corollary 3.2 describes log-ratios, but an embedding uses the affinity as a Markov chain: the mass P_{ij} that x_i sends to x_j. We now compute where that chain moves mass, as the bandwidth vanishes.

Setting. Let $p \in C^2(\mathbb{T}^d)$ be a density on the flat torus with $p \geq p_{\min} > 0$. Let $\omega \in C^2(\mathbb{T}^d; \mathbb{R}^d)$ with $\sup \|\omega\| < 1$, and let $\sigma \in C^2(\mathbb{T}^d)$ be a positive relative bandwidth. The torus avoids boundaries and curvature terms; see the remark after Corollary 4.3 for the harmonic part it introduces. Let $g : [0, \infty) \to (0, \infty)$ be a decreasing profile. The two cases of interest are $g(t) = e^{-t}$ (linear cost) and $g(t) = e^{-t^2/2}$ (squared cost). With $\Delta = y - x$ the torus displacement, the frozen Randers kernel is

$$k_h(x, y) = g\left(\frac{F_x(\Delta)}{h\,\sigma(x)}\right), \qquad F_x(v) = \|v\| + \omega(x)^\top v. \tag{5}$$

- The *balanced chain* is $T_h(x, dy) = u_h(x)\, k_h(x, y)\, v_h(y)\, p(y)\, dy$, with u_h, v_h chosen so that $T_h(x, \cdot)$ is a probability and p is stationary: $\int p(x) T_h(x, dy)\, dx = p(y)\, dy$. This is the static Schrödinger problem with both marginals equal to p (Léonard, 2014).
- The *row-stochastic chain* is $T_h^{\mathrm{RS}}(x, dy) \propto k_h(x, y)\, p(y)\, dy$.

The *transport field*, in units of the bandwidth, is $b_h(x) = h^{-1} \int (y - x)\, T_h(x, dy)$.

Tilted log-partition. For $\omega \in \mathbb{R}^d$ with $\|\omega\| < 1$, define on its open convex domain

$$\Lambda(\omega, \zeta) = \log \int_{\mathbb{R}^d} g\big(F_\omega(v)\big)\, e^{\zeta^\top v}\, dv, \qquad M = \nabla_\zeta \Lambda. \tag{6}$$

$M(\omega, \zeta)$ is the mean of the profile tilted by $e^{\zeta^\top v}$. $\Lambda(\omega, \cdot)$ is a log-Laplace transform of a non-degenerate measure, hence strictly convex.

Assumption 4.1 (Small-noise potentials)**.** $\log v_h = \Psi/h + R_h$, with $\Psi \in C^2(\mathbb{T}^d)$, and R_h bounded in C^1 uniformly in h. Moreover $\sigma \nabla \Psi(x)$ lies in the domain of $\Lambda(\omega(x), \cdot)$ for every x.

This is the usual WKB behaviour of Schrödinger potentials (Léonard, 2014). It holds exactly for gradient drifts with the linear profile: there $v_h = e^{\phi/h}$ times the symmetric scaling.

Theorem 4.2 (Balanced transport is p-solenoidal; formal)**.** *Under Assumption 4.1, $b_h \to b$ uniformly, where*

$$b(x) = \sigma(x)\, M\big(\omega(x),\, \sigma(x) \nabla \Psi(x)\big), \qquad \operatorname{div}(p\, b) = 0. \tag{7}$$

The potential Ψ is, up to a constant, the unique minimiser of the strictly convex functional

$$J(\Psi) = \int_{\mathbb{T}^d} p(x)\, \Lambda\big(\omega(x), \sigma(x) \nabla \Psi(x)\big)\, dx. \tag{8}$$

For the row-stochastic chain, by contrast, $b_h^{\mathrm{RS}} \to \sigma\, M(\omega, 0) = \sigma\, \kappa_g\, c_F$. Here $c_F(x)$ is the centroid of the Randers unit ball $\{F_x \leq 1\}$, and $\kappa_g = \int t^{d+1} |g'|\, dt \,/ \int t^d |g'|\, dt$.

The proof sketch, in three steps (local profile, stationarity, variational form), is in Appendix D.

Theorem 4.2 is the Sinkhorn counterpart of the antisymmetric limits of Perrault-Joncas & Meilă (2011) and Debaussart-Joniec et al. (2026). Their row- and θ-normalised operators converge to transport along the centroid field. Balancing adds the tilt $\sigma \nabla \Psi$, chosen to cancel exactly the part of that transport which would change the density.

4.1. The Laplace–Randers kernel in closed form

For $g(t) = e^{-t}$ the tilted log-partition is explicit. With $\eta = \omega - \zeta$,

$$\Lambda(\omega, \zeta) = \log c_d - \tfrac{d+1}{2} \log\big(1 - \|\eta\|^2\big), \qquad \|\eta\| < 1. \tag{9}$$

This follows from $\int e^{-\|v\| - \eta^\top v} dv = c_d (1 - \|\eta\|^2)^{-(d+1)/2}$, which can be checked by integrating in polar coordinates. The Randers unit ball is an ellipsoid with centroid $c_F = -\omega/(1 - \|\omega\|^2)$, and $\kappa_g = d + 1$.

Corollary 4.3 (Laplace–Randers, global bandwidth)**.** *Let $g(t) = e^{-t}$ and $\sigma \equiv 1$. Then*

$$b = -(d+1)\, \frac{\omega - \nabla \Psi}{1 - \|\omega - \nabla \Psi\|^2},$$

$$\Psi \in \arg\min_{\Psi} \; -\int p\, \log\big(1 - \|\omega - \nabla \Psi\|^2\big),$$

whereas $b^{\mathrm{RS}} = -(d+1)\, \omega/(1 - \|\omega\|^2)$. Moreover:

1. *$b \equiv 0$ if and only if ω is a gradient on $\mathbb{T}^d$. This is the continuum version of Corollary 3.3, valid for every density p.*

2. *At small drift,* $b = -(d+1)\,\Pi_p\,\omega + O(\|\omega\|^3)$. *Here* $\omega = \nabla\Psi_0 + \Pi_p\omega$ *is the* p*-weighted Helmholtz decomposition, with* $\operatorname{div}(p\,\Pi_p\omega) = 0$.

The proof is in Appendix D.

So a balanced Laplace–Randers affinity is, to first order, a *density-weighted Leray projector*: it moves mass along the p-solenoidal part of $-\omega$, and only that. The row-stochastic chain moves mass along all of $-\omega$. Note that the weight is the sampling density. Whether a drift counts as circulating depends on how the points are distributed, not only on the geometry.

On $\mathbb{T}^d$, constant fields are not gradients of periodic functions, so Π_p also keeps a harmonic component; on $\mathbb{R}^d$ with a confining density, a constant field is the gradient of $c^\top x$ and is invisible, as in Corollary 3.3.

5. Leakage: what practical affinities let through

Neighbour embeddings rarely use a linear cost with one global bandwidth. SNE-type affinities use a *squared* cost and *per-row* bandwidths calibrated to a perplexity. Both break the gauge.

Finite samples. Both effects multiply the gradient flow G entrywise by a symmetric weight m. If the effective log-kernel has antisymmetric part $-(m \odot G)$, then for any constant c we have $\operatorname{cyc}_W(m \odot G) = \operatorname{cyc}_W((m-c) \odot G)$, since $cG \in \mathcal{G}$. Therefore

$$\big\|\operatorname{cyc}_W(m \odot G)\big\|_W \;\le\; \inf_c \big\|(m-c) \odot G\big\|_W. \qquad (10)$$

The leaked asymmetry is controlled by the *heterogeneity* of m over the edges that carry weight. The two cases are:

- *Squared cost.* With $(D+G)^2 = D^2+G^2+2D\odot G$, the weight is $m = 2D/\varepsilon$, the distance itself. The leak is set by the spread of neighbour distances, which shrinks as $1/\sqrt{2d}$ for Gaussian neighbourhoods in intrinsic dimension d.
- *Per-row bandwidths.* Write $\beta_i = 1/\varepsilon_i$. The antisymmetric part of $\beta_i C_{ij}$ is $\frac{1}{2}(\beta_i-\beta_j)S_{ij} + \frac{1}{2}(\beta_i+\beta_j)A_{ij}$. The first term is the familiar drift-free asymmetry of SNE conditionals. The drift enters with weight $m_{ij} = \frac{1}{2}(\beta_i+\beta_j)$, so it leaks in proportion to how much bandwidths vary across edges.

In both cases the leak in $\log(P_{ij}/P_{ji})$ is of first order in the drift. The continuum limit separates the two effects sharply.

Corollary 5.1 (Leakage orders)**.** *Under Assumption 4.1, let* $\omega = \nabla\phi$ *with* $\|\nabla\phi\| = O(\rho)$.

1. ***Squared cost, global bandwidth*** *(*$g = e^{-t^2/2}$*,* $\sigma \equiv 1$*). Then* $b = O(\rho^3)$*, and* b *is generically non-zero.*
2. ***Linear cost, per-row bandwidth*** *(*$g = e^{-t}$*). Then* $b = -(d+1)\,\sigma^2\,\Pi_{p\sigma^2}(\nabla\phi/\sigma) + O(\rho^3)$*. This is first order in* ρ*, and it vanishes iff* $\nabla\phi/\sigma$ *is a gradient, i.e. iff* $d\sigma \wedge d\phi = 0$.

Proof sketch. (1) The map M is odd: $M(-\omega,-\zeta) = -M(\omega,\zeta)$. Linearising around 0 for the standard Gaussian profile gives $M(\omega,\zeta) = \zeta - s_0\,\omega + O(3)$, with $s_0 = 2\,\mathbb{E}\|v\|^3/d$ and $v \sim \mathcal{N}(0, I_d)$. The linear part of the drift is therefore the gauge $\zeta = s_0\nabla\phi$, which is a gradient. What remains is the p-solenoidal part of a cubic remainder. (2) Here Λ depends on $\omega - \zeta$ only. So $b = \sigma M(\omega, \sigma\nabla\Psi) \approx -(d+1)\,\sigma^2(\omega/\sigma - \nabla\Psi)$. Stationarity $\operatorname{div}(p\,b) = 0$ is then the weighted Helmholtz problem with weight $p\sigma^2$. □

Perplexity calibration makes σ scale roughly like $p^{-1/d}$. Under per-row bandwidths, a gradient drift is thus invisible only when the density is constant on the level sets of its potential. This is the dominant leak at small drift. In the limit, the squared cost leaks less in transport even though it leaks comparably in log-ratios: what it adds is an odd, skewed asymmetry whose mean displacement cancels at first order. At finite sample sizes this cancellation is incomplete (Section 6).

6. Numerical verification

We check each statement on synthetic data where the Helmholtz parts of the drift are known exactly. Appendix C gives the protocol. All couplings are computed in float64 and balanced to 10^{-11}.

Finite-sample identities. On 400 points in $\mathbb{R}^5$ with a generic cost:

- adding a gradient flow G to the cost changes the balanced coupling by 2×10^{-14} in relative ℓ_1 norm, and the row-stochastic one by 0.11 (Theorem 3.1);
- the cyclic part of $\log(P/P^\top)$ equals $-\frac{2}{\varepsilon}\operatorname{cyc}(A)$ to a relative error of 5×10^{-15}, although the log-ratio also has a large gradient part, 31% of its norm (Corollary 3.2).

Transport fields. On the flat torus $\mathbb{T}^2$ with a non-uniform density, we use a drift $\omega = \rho(\nabla\phi + w)$ whose p-weighted decomposition is known by construction: $\operatorname{div}(p\,w) = 0$. Figure 1 shows the result for the linear Randers cost with a global bandwidth. Corollary 4.3 predicts that the balanced chain moves along $-3\rho\,w$ and the row-stochastic chain along $-3\rho(\nabla\phi + w)$.

Table 1. Bandwidth sweep at a fixed number of neighbours per kernel, $\rho = 0.3$: slope of measured on predicted transport (1 is exact), and the gradient leak under per-row bandwidths with the cosine of its Corollary 5.1 prediction.

h	n	slope, Sk	slope, RS	per-row cos
0.070	980	0.69	0.64	0.95
0.050	1920	0.80	0.78	0.96
0.040	3000	0.84	0.86	0.96
0.035	3918	0.86	0.89	0.97

- **Direction.** The balanced transport has cosine 0.99 with the predicted solenoidal field, and -0.001 with the gradient part. The row-stochastic transport has cosine 0.98 with the whole drift. This holds at both drift strengths tested.
- **Magnitude.** The regression slope of measured on predicted transport approaches 1 as the bandwidth shrinks (Table 1). The deficit roughly halves when h halves, as expected from an $O(h)$ correction.
- **Solver check.** Computing the prediction with a finite-volume weighted Helmholtz solver instead of the known w gives the same cosine and slope.

Leakage. Figure 2 compares a gradient drift with a solenoidal drift of the same strength, under the three costs.

- **Linear cost, global bandwidth.** The gradient drift produces no balanced transport, below 4×10^{-11} for every ρ, at the level of the Sinkhorn tolerance (Corollaries 3.3 and 4.3).
- **Linear cost, per-row bandwidth.** The gradient drift leaks at first order in ρ, and strongly: about 80% of the transport produced by a solenoidal drift of the same strength. The leak matches Corollary 5.1(2) with cosine 0.96 to 0.97 and slope 0.92. We obtain it from the measured per-row bandwidths through a weighted Helmholtz solve.
- **Squared cost, global bandwidth.** The leak is also linear in ρ over the range we can simulate, at about 14% of the transport of a solenoidal drift under the same cost. It shrinks when the sample size grows at fixed bandwidth (0.020, 0.017, 0.013 at $\rho = 0.4$ for $n = 980, 1960, 3920$, $h = 0.07$), so it is at least partly a finite-sample effect. We do not reach the third-order regime of Corollary 5.1(1), and we leave its numerical confirmation open.

7. Consequences for real velocity data

The theory predicts when a balanced affinity helps. What matters is how a velocity field splits into a gradient part and a circulating part. We decompose the edge flow $f_{ij} = \frac{1}{2}(v_i + v_j)^\top (x_j - x_i)$ on a 30-nearest-neighbour graph in 30 principal components, following Jiang et al. (2011). The null model permutes the centred velocities among cells.

Table 2. Hodge content of real RNA-velocity fields: share of $\|f\|^2$ carried by the gradient part, its permutation null, and the part that carries the ground-truth progress. Gastrulation: $n = 30{,}000$ cells, E6.5–E8.5 (Pijuan-Sala et al., 2019). RPE1: $n = 2{,}793$ cells with FUCCI phase (Battich et al., 2020), velocity from dynamo (Qiu et al., 2022).

velocity	gradient	null	progress carried by
gastrulation, direct	0.93	0.33	gradient
gastrulation, scVelo	0.78	0.31	gradient
RPE1, direct	0.67	0.21	non-gradient
RPE1, dynamo kernel	0.29	0.24	mostly non-gradient

Table 3. Balanced (Sk) versus row-stochastic (RS) Randers affinities built from the same velocity: squared cost, per-row perplexity $\sqrt{n}$, dense. "cos" is the mean cosine between each cell's shift in outgoing displacement, relative to the symmetric affinity, and the gradient or non-gradient Hodge part of the velocity. "progress" is the ground-truth progress after 10 steps of the walk, minus that of the symmetric walk: pseudotime units for pancreas, fractions of a cycle for RPE1. The linear cost gives the same picture.

data	norm.	cos grad.	cos rest	progress
pancreas	RS	0.56	0.21	0.167
	Sk	0.38	0.26	0.000
RPE1, direct	RS	0.56	0.14	0.060
	Sk	0.01	0.33	0.082
RPE1, kernel	RS	0.34	0.36	0.311
	Sk	0.12	0.37	0.241

Table 2 shows two opposite regimes.

- **Developmental trajectories are gradient-dominated.** On gastrulation, a walk along the gradient part alone recovers the whole forward-in-time signal. The remainder carries none of it. A large share of the velocity is also a vector common to all cells, which Corollary 3.3 makes invisible to any balanced affinity.
- **The cell cycle circulates.** On RPE1, the gradient part of the direct velocity makes walks go slightly *backwards* in phase. The non-gradient part carries the progress around the cycle; for the kernel velocity it carries most of it over long walks. The velocity circulates around closed loops through the phase bins far above the null (z between 5 and 18).

Balanced versus row-stochastic affinities. Table 3 builds both affinities from the same velocity, with the per-row perplexity calibration of practical neighbour embeddings.

Three observations follow the theory.

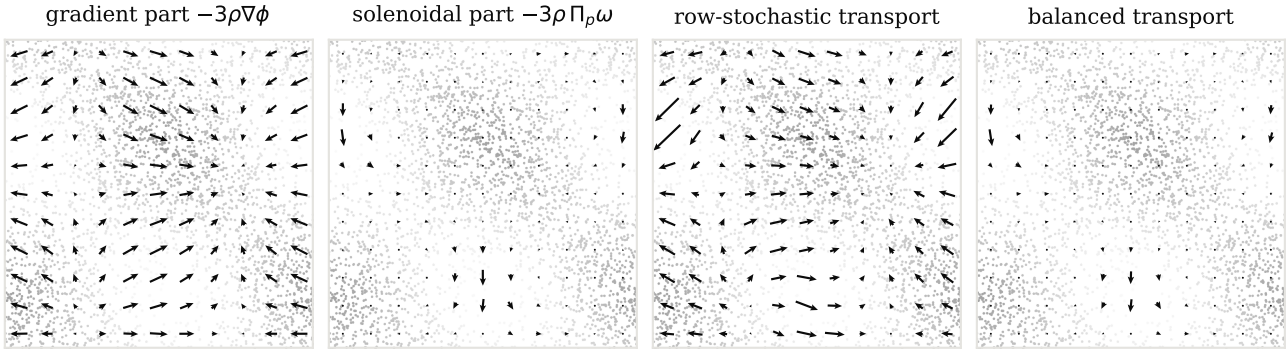


Figure 1. Transport on the flat torus, drift $\omega = \rho(\nabla\phi + w)$ with $\mathrm{div}(p\,w) = 0$, $\rho = 0.3$, linear Randers cost, global bandwidth ($n = 3000$, $h = 0.04$). Arrows are averages over grid cells, all panels on one scale; grey dots are the samples, darker where p is higher. The row-stochastic chain moves along the whole drift. The balanced chain moves along its p-solenoidal part only, as predicted by Corollary 4.3.

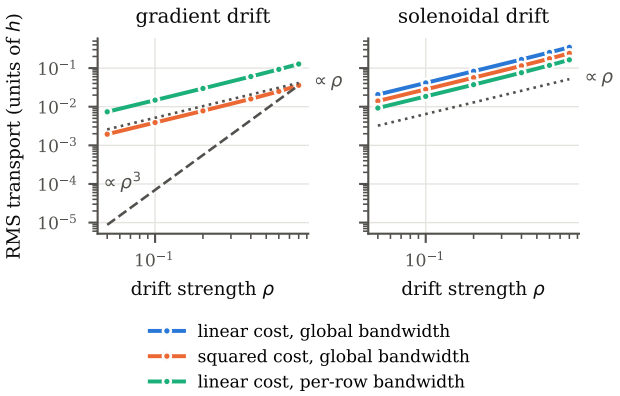


Figure 2. RMS balanced transport against drift strength on the torus ($n = 3000$, $h = 0.04$). Left: a gradient drift. With the linear cost and a global bandwidth it stays below 4×10^{-11} for every ρ, outside the plotted range. Right: a solenoidal drift of the same strength, for reference.

- **No net progress along a potential.** On pancreas, the balanced walk's mean pseudotime gain is zero to numerical precision, as Proposition 3.4 requires. The row-stochastic walk gains steadily.

- **Circulation survives balancing.** On RPE1, the balanced walk keeps the progress around the cycle. With the direct velocity it progresses further than the row-stochastic walk, whose extra sensitivity to the gradient part pushes it backwards in phase.

- **The balanced shift drops the gradient part.** On RPE1 its correlation with the gradient part falls to about zero for the direct velocity, while its correlation with the remainder rises. On pancreas the balanced shift still has a sizeable gradient correlation. This is the first-order leak of per-row bandwidths (Corollary 5.1). By Proposition 3.4 it must be compensated, and it is: per pseudotime decile, the balanced chain moves cells forward in the first seven and backwards in the last three. Its largest step, -0.076, is in the terminal decile. The row-stochastic chain moves forward in every decile.

Consequence for embeddings. These numbers explain an observation from fitting Randers neighbour embeddings on pancreas (Bastidas-Ponce et al., 2019). With a row-stochastic input affinity, the fitted flow agreed with the true velocity at cosine 0.80 and with scVelo's projection at 0.76. With a balanced one, these fell to 0.69 and 0.54. The balanced direction was attenuated rather than erased, as Corollary 5.1 predicts: per-row bandwidths and the squared cost leak part of the gradient drift back.

8. Related work

Entropic and balanced affinities. Perplexity-calibrated affinities come from SNE (Hinton & Roweis, 2002; van der Maaten & Hinton, 2008; Vladymyrov & Carreira-Perpiñán, 2013). Doubly stochastic normalisation was proposed for spectral clustering (Zass & Shashua, 2006). It has well-understood continuum limits for symmetric kernels (Marshall & Coifman, 2019) and is robust to heteroskedastic noise (Landa et al., 2021). SNEkhorn brought it to neighbour embeddings with symmetry imposed (Van Assel et al., 2023). All of these use symmetric costs or couplings; we balance an asymmetric cost without imposing symmetry.

Directed kernels and their limits. Perrault-Joncas & Meilă (2011) split the continuum limit of a directed kernel into a Laplacian and a drift. Debaussart-Joniec et al. (2026) build Finsler kernels from directed distances and prove that the antisymmetric part of the θ-normalised operator converges to transport along the centroid field. Theorem 4.2 is the balanced counterpart, and it shows that the choice

of normalisation decides which Helmholtz component of a drift is visible.

Hodge decompositions of flows. Combinatorial Hodge theory decomposes edge flows into gradient, curl and harmonic parts (Jiang et al., 2011; Lim, 2020). Vector-field reconstructions of single-cell dynamics analyse the same split (Qiu et al., 2022). Weinreb et al. (2018) show that snapshots cannot identify the rotational part of a dynamics; Section 3 shows the complementary blindness of balanced affinities.

9. Discussion and limitations

What changes in practice. For drifts dominated by their gradient part, such as developmental trajectories, a balanced input affinity discards most of the direction, while a row-stochastic one keeps it as a marginal tilt; an exact potential cost then gives a reweighted symmetric affinity. For circulating drifts, such as the cell cycle, balancing isolates the circulation. Arrows drawn from a balanced embedding show a p-weighted Helmholtz projection of the velocity, not the velocity itself.

Limitations. Theorem 4.2 is formal: it rests on the small-noise ansatz for Sinkhorn potentials, and a rigorous version would combine the consistency arguments of Debaussart-Joniec et al. (2026) with quantitative stability of Schrödinger potentials. It covers the first-order transport on a flat torus with the frozen kernel, without curvature, boundaries or the second-order Laplacian part. The third-order leak of the squared cost is not observed numerically: at our sample sizes it leaks at first order, with a constant that decreases with n. The *output* kernel of an embedding, where the same gauge acts on the fitted drift, is left to future work.

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A. Computing Randers distances

Low dimension: eikonal solvers. For a fixed source x_0, $u = d_F(x_0 \to \cdot)$ is the viscosity solution of the eikonal equation $F_x^*(\nabla u) = 1$, $u(x_0) = 0$, where $F_x^*(p) = \sup_{F_x(v)\le 1} p^\top v$ is the dual norm. For a Randers norm the dual is again of Randers type, given in closed form by Zermelo's navigation (Bao et al., 2000). On a grid the equation is solved by ordered-upwind methods (Sethian & Vladimirsky, 2003) or by fast marching with adaptive stencils, which handle anisotropic and asymmetric metrics (Mirebeau, 2014). Each solve costs $O(N \log N)$ on N grid nodes, and the full cost matrix needs one solve per source, so n solves. This is practical in $d = 2$ or 3.

High dimension. A grid with m nodes per axis has m^d nodes, far out of reach for the 10 to 50 principal components used on single-cell data. Graph substitutes replace paths by chains of nearest-neighbour edges, each carrying a local Randers length, and run Dijkstra's algorithm on the directed graph, as Isomap does for Euclidean geodesics (Tenenbaum et al., 2000). Heat-kernel proxies replace the infimum by a small-time limit of a diffusion. Both converge only as the sampling densifies around the optimal paths, which in high dimension takes exponentially many points, and both are biased toward paths through dense regions.

The drift is only sampled. A velocity is estimated at each point x_i and nowhere else. The infimum ranges over paths through regions where nothing is measured, so the value of d_F depends on how ω is extended between samples, by interpolation or by a fitted network. That is a modelling choice the data do not support.

Error of the closed forms. The straight segment $\gamma(t) = x_i + t\Delta$, $\Delta = x_j - x_i$, gives the upper bound

$$d_F(x_i \to x_j) \le \|\Delta\| + \int_0^1 \omega(x_i + t\Delta)^\top \Delta \, dt.$$

Perturb the segment by $\eta(t)$ with $\eta(0) = \eta(1) = 0$ and $\|\eta\| \le h$. The length grows by order h^2/r, with $r = \|\Delta\|$. The first variation of the drift term is $\int_0^1 \eta^\top (\nabla\omega - \nabla\omega^\top)\dot\gamma \, dt$, of order $\|\nabla\omega - \nabla\omega^\top\| \, r\, h$; only the curl of ω enters it. Optimising over h gives a gain of order $\|\nabla\omega - \nabla\omega^\top\|^2 r^3$, which vanishes for gradient drifts. The frozen form is the left-endpoint rule for the line integral, with error $O(\|\nabla\omega\| r^2)$; the averaged form is the trapezoid rule, with error $O(\|\nabla^2\omega\| r^3)$.

B. Computations omitted from the main text

The Laplace–Randers partition function. For $\|\eta\| < 1$ we claim $Z_d(\eta) = \int_{\mathbb{R}^d} e^{-\|v\| - \eta^\top v} dv = c_d \, (1 - \|\eta\|^2)^{-(d+1)/2}$. By rotation invariance Z_d depends on $s = \|\eta\|$ only. In polar coordinates, $Z_d(s) = \int_0^\infty r^{d-1} e^{-r} \, \Phi_d(sr) \, dr$, where $\Phi_d(t) = \int_{S^{d-1}} e^{-t\theta_1} d\theta$. Three cases check the claim directly:

- $d = 1$: $\int e^{-|v| - sv} dv = \frac{1}{1-s} + \frac{1}{1+s} = \frac{2}{1-s^2}$.
- $d = 2$: $Z_2 = 2\pi \int_0^\infty r e^{-r} I_0(sr) \, dr = 2\pi (1 - s^2)^{-3/2}$, the Laplace transform of $r I_0(sr)$ evaluated at 1.
- $d = 3$: $Z_3 = \frac{4\pi}{s} \int_0^\infty r e^{-r} \sinh(sr) \, dr = 8\pi (1 - s^2)^{-2}$.

For general d, $\Phi_d(t) = c'_d \, t^{-(d-2)/2} I_{(d-2)/2}(t)$, and the Laplace transform of $r^{d/2} I_{(d-2)/2}(sr)$ at 1 is proportional to $s^{(d-2)/2}(1 - s^2)^{-(d+1)/2}$. This gives the claim. Differentiating (9) in ζ gives $M(\omega, \zeta) = -(d+1)\, \eta / (1 - \|\eta\|^2)$ with $\eta = \omega - \zeta$.

The Randers unit ball. $\{v : \|v\| + \omega^\top v \le 1\}$ is the set where $\|v\|^2 \le (1 - \omega^\top v)^2$ and $\omega^\top v \le 1$. For $\|\omega\| < 1$ this is an ellipsoid, symmetric about the axis ω. Along that axis, $v = t\, \omega / \|\omega\|$ with $t \in [-1/(1 - \|\omega\|), \, 1/(1 + \|\omega\|)]$. Its centre is therefore $c_F = -\omega / (1 - \|\omega\|^2)$. With $\kappa_g = d + 1$ for $g = e^{-t}$, this agrees with $M(\omega, 0)$ above.

Linearisation for the Gaussian profile. For $g(t) = e^{-t^2/2}$, $F_\omega(v)^2 = \|v\|^2 + 2\|v\| \, \omega^\top v + (\omega^\top v)^2$. To first order in (ω, ζ) the tilted density is $\mathcal{N}(0, I_d) \cdot \big(1 + \zeta^\top v - 2\|v\| \, \omega^\top v\big)$. Its mean is

$$\zeta - 2\, \mathbb{E}\big[\|v\| \, vv^\top\big] \, \omega = \zeta - \frac{2\, \mathbb{E}\|v\|^3}{d} \, \omega.$$

So $s_0 = 2\mathbb{E}\|v\|^3/d$; in $d = 2$, $\mathbb{E}\|v\|^3 = 3\sqrt{\pi/2}$ and $s_0 \approx 3.76$. The map $(\omega, \zeta) \mapsto M$ is odd, so the next term is cubic.

Per-row bandwidths in the discrete prediction. Section 6 compares measured transport with Corollary 5.1(2). It uses the discrete form of that statement. With kernel $\exp(-\beta_i C_{ij})$ and $\beta(x) = 1/(h\sigma(x))$, the predicted mean displacement is $-(d+1)\,u/\beta^2$. Here $u = \beta\omega - \nabla\Psi$ is the solenoidal part of $\beta\omega$ in $L^2(p/\beta^2)$. We compute it by finite volumes on a 128×128 periodic grid, after smoothing $1/\beta_i$ onto the grid with a Gaussian of width 0.04.

C. Experimental protocol

All experiments run in float64 with log-domain Sinkhorn, stopped when every row sum is within 10^{-11} relative error. Code: `experiments/theory/gauge_checks.py`.

Finite-sample identities. $n = 400$ Gaussian points in $\mathbb{R}^5$, $\varepsilon = 0.3$, Euclidean S. The gradient part uses $\phi = 0.2(\tanh x_1 + 0.3x_2^2)$; the generic antisymmetric part is a random Gaussian matrix, antisymmetrised, of scale 0.1.

Torus. Points are sampled on $\mathbb{T}^2$ by rejection from $p(x) \propto 1 + 0.6\cos(2\pi x_1)\cos(2\pi(x_2 - 0.15)) + 0.25\sin(2\pi(x_1 + x_2))$.

- Gradient part: $\phi = \big(\sin 2\pi x_1 + \frac{1}{2}\sin 2\pi(x_2 + 0.2)\big)/2\pi$.
- Solenoidal part: $w = p^{-1}\nabla^\perp\chi$ with $\chi = \sin(2\pi x_1)\sin(2\pi x_2)/2\pi$, so that $\operatorname{div}(p\,w) = 0$ by construction. Hence $\Pi_p(\nabla\phi + w) = w$ exactly.

Both fields are scaled to maximum norm 1 and multiplied by ρ. The cost is the linear Randers cost with the exact potential term and the averaged form for w: $C_{ij} = \|\Delta_{ij}\| + \rho(\phi_j - \phi_i) + \rho\,\bar{w}_{ij}^\top\Delta_{ij}$, with torus displacements. The transport field is $b_i = h^{-1}\sum_j P_{ij}\Delta_{ij}/\sum_j P_{ij}$. The same quantity for the drift-free kernel is subtracted, to remove the $O(h)$ density term. Per-row bandwidths are calibrated by bisection to perplexity 30.

D. Proof sketch of Theorem 4.2

Proof sketch of Theorem 4.2. Step 1: local profile. Substitute $y = x + h\sigma(x)v$. Then $v_h(y)/v_h(x) = \exp\big(\sigma\nabla\Psi(x)^\top v + O(h\|v\|^2 + h\|v\|)\big)$ and $p(y)/p(x) = 1 + O(h\|v\|)$. So the law of v under $T_h(x,\cdot)$ converges to the normalised density $g(F_{\omega(x)}(v))\,e^{\sigma\nabla\Psi^\top v}$. Its mean is $M(\omega, \sigma\nabla\Psi)$, and the displacement is $h\sigma v$, which gives (7). Tails are controlled because the tilt stays inside the domain of Λ.

Step 2: stationarity. For $f \in C^2$, stationarity of p gives $\int p(x)\int \big(f(y) - f(x)\big)T_h(x, dy)\,dx = 0$. A Taylor expansion turns this into $h\int p\,b_h^\top\nabla f + O(h^2) = 0$. Hence $\int p\,b^\top\nabla f = 0$ for all f, i.e. $\operatorname{div}(p\,b) = 0$.

Step 3: variational form. The first variation of (8) is $\delta J = \int p\,\sigma\,\nabla_\zeta\Lambda^\top\nabla\delta\Psi = -\int \operatorname{div}(p\,b)\,\delta\Psi$. So Step 2 is the Euler–Lagrange equation of J. Strict convexity of Λ in ζ makes the critical point unique up to constants.

Row-stochastic case. There is no tilt, so the mean is $M(\omega, 0)$. Write $g(F) = \int_0^\infty |g'(t)|\,\mathbf{1}\{F \le t\}\,dt$ and use the 1-homogeneity of F: $\int v\,\mathbf{1}\{F(v) \le t\}\,dv = t^{d+1}\int_{\{F\le 1\}} v\,dv$. The ratio of the first moment to the mass gives $\kappa_g c_F$. □

Proof of Corollary 4.3. (1) The integrand $-\log(1 - t^2)$ is non-negative and vanishes only at $t = 0$. If $\omega = \nabla\phi$, then $\Psi = \phi$ attains the minimum value 0 and $b = 0$. Conversely, $b = 0$ forces $\omega = \nabla\Psi$. (2) Since $-\log(1 - t^2) = t^2 + O(t^4)$, Ψ minimises $\int p\,\|\omega - \nabla\Psi\|^2$ to leading order, which is the weighted L^2 projection onto gradients. □